
HEMOVISION

Contactless Vital-Sign Estimation from Facial Video

Remote Photoplethysmography (rPPG) —

Vital signs usually need a wire, a cuff, or a clip



Traditional monitoring

- Requires physical contact — cuffs, clips, wired sensors
- Equipment isn't always available at the point of care
- Slows down outpatient triage when volumes are high



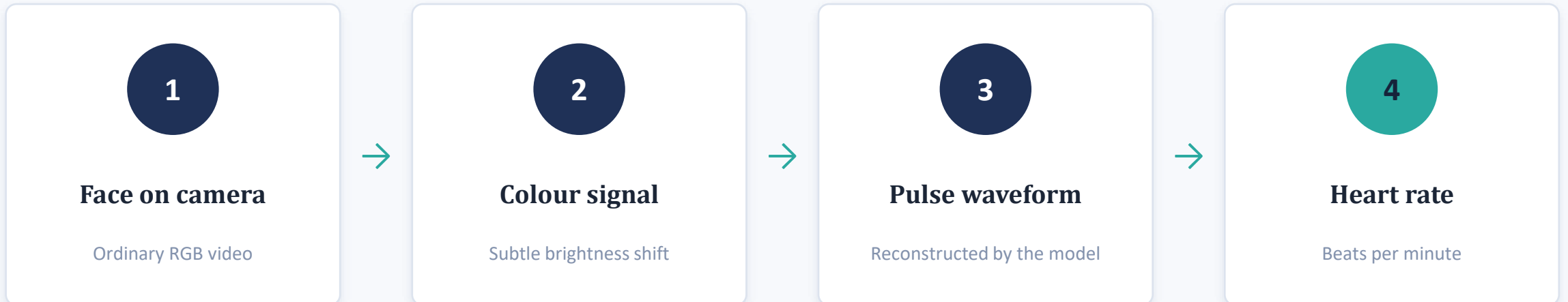
Our approach

- Just a face, on ordinary video — no sensor to attach
- Works with a standard camera — webcam or phone
- Deep learning reconstructs the pulse from color alone

THE IDEA

Every heartbeat, invisibly, changes your skin colour

With each pulse of blood, the light absorbed by facial skin shifts by a fraction the eye can never see. A camera, watching closely enough over time, can.



THE DATA

MCD-rPPG — a clinically-grounded benchmark

Every video is paired with real medical-device measurements, so every prediction can be checked against the truth.

600

subjects

3

camera types

2

conditions
(resting / post-exercise)

13

clinical ground-truth
measurements

Reference signals include synchronized ECG and clinical PPG — not self-reported or estimated values.

From raw video to a heart-rate estimate



The model never guesses a number directly — it reconstructs the pulse wave itself, then measures its frequency. That distinction turns out to matter a great deal (see Slide 9).

Heart rate: accurate on subjects never seen before

9.0 BPM

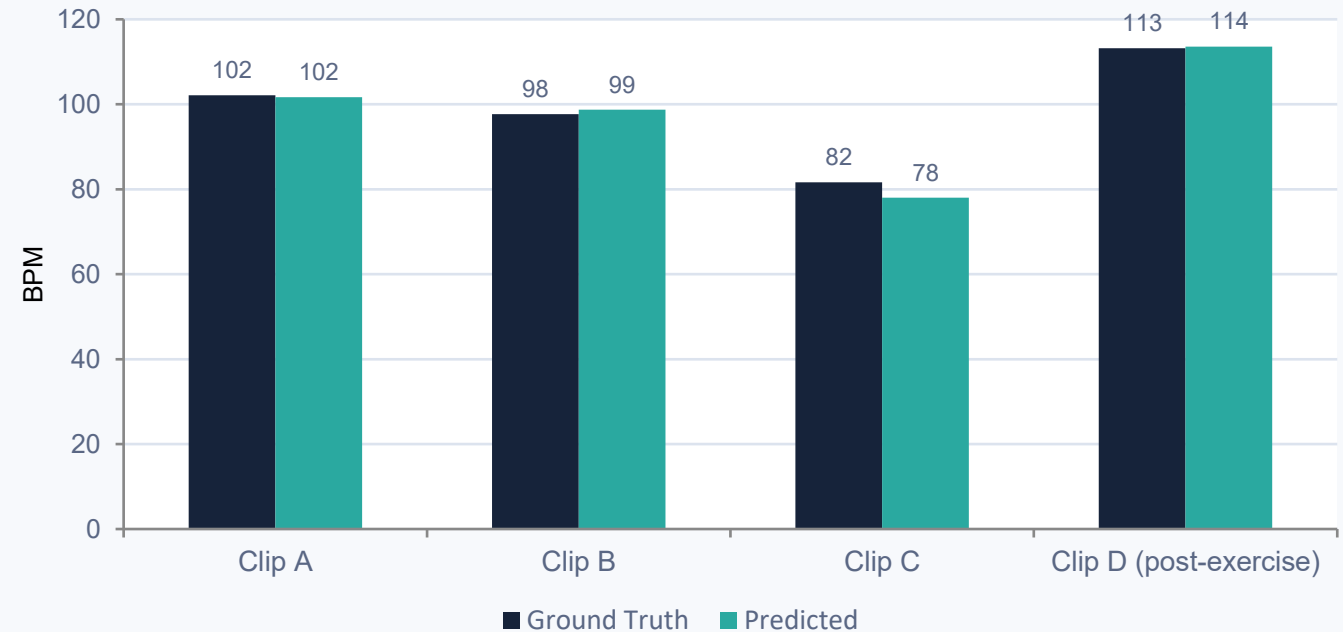
average error, held-out test set (90 subjects)

$r \approx 0.28$

correlation with true heart rate

Derived from the reconstructed waveform — not a direct guess.

Verified demo clips — ground truth vs. predicted (BPM)



Every strange result became a bug to explain

1

Ground-truth double-counting

The reference heart-rate detector mistook extra heartbeat spikes on post-exercise recordings, doubling the recorded rate for some subjects. Fixing it cut baseline error by more than half.

2

Frequency-resolution rounding

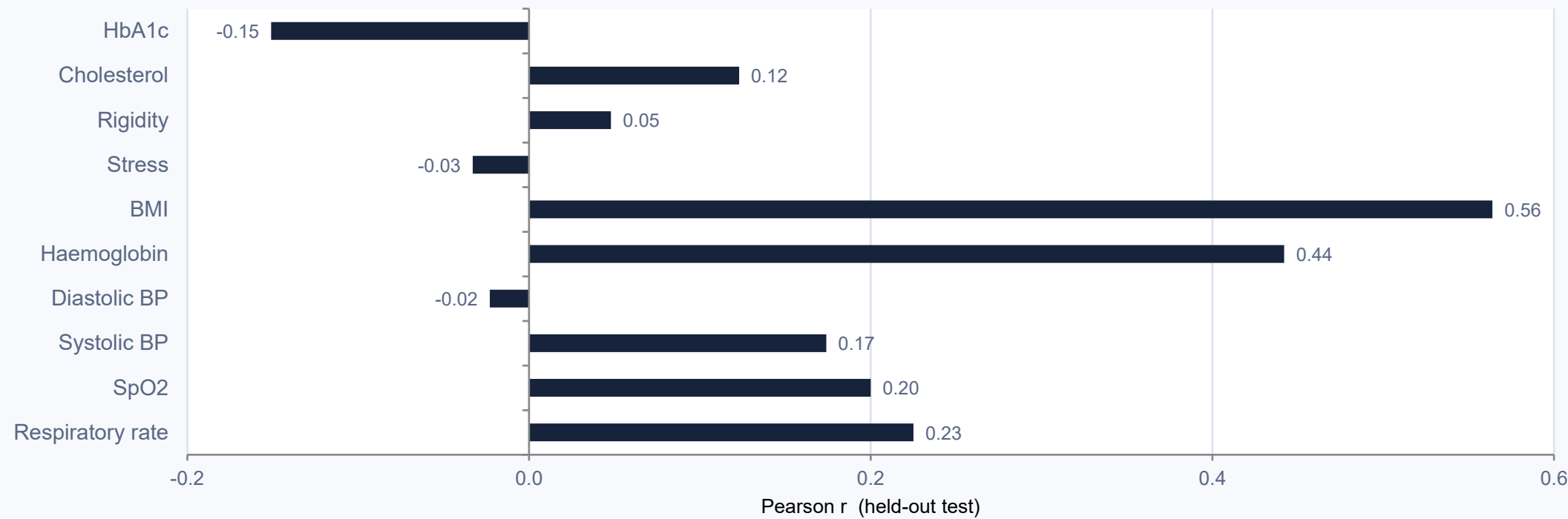
An unpadded frequency calculation forced every estimate onto a coarse 7.5 BPM grid. Identified via a failing precision self-test and corrected.

3

Unstable checkpoint selection

One noisy, unrepresentative training step was being silently saved as “best.” Added stability checks so training can't be fooled by a lucky fluke.

Beyond heart rate: what we tested, and what we found

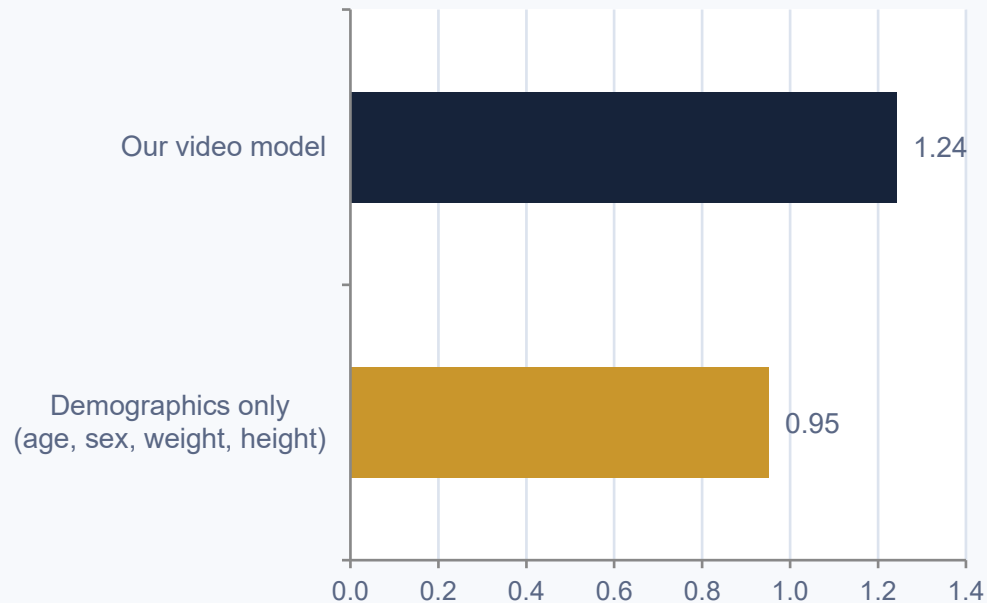


None of these ten measurements reliably beat a simple “guess the average” baseline ($r < 0.3$) — confirmed after five independent attempts to fix the model architecture.

CASE STUDY

Hemoglobin — a controlled experiment, not a guess

Prediction error (g/dL)



Demographics alone predict hemoglobin better than our video model.

A controlled test: we compared video-based predictions against a simple estimate built only from age, sex, weight, and height — no video at all.

Repeated label values in the dataset suggest the model may be recognising the person from their face — not measuring blood chemistry.

WHY

Some signals need light our camera can't see

True oxygen and hemoglobin measurement relies on comparing how blood absorbs different wavelengths of light — including infrared. An ordinary camera only records three overlapping visible-light channels.

Pulse oximeter

Red + infrared wavelengths

Two distinct wavelengths let it tell oxygenated and deoxygenated blood apart directly.

RGB camera

Red + green + blue only

No infrared channel — a physical sensor limitation, not a modelling shortfall.

WHAT'S NEXT

Three concrete directions, each grounded in the evidence



Blood pressure

Analyse the pulse wave's timing and shape (transit time / morphology) instead of a single averaged value — a technique this project hasn't yet tried.



Stress, via HRV

Heart-rate variability, computed from the waveform that already works, offers an indirect but literature-backed path to stress estimation.



Multi-wavelength hardware

Pursuing a research collaboration for infrared/multi-wavelength imaging — the missing piece for hemoglobin and oxygen estimation.

Summary

A working system — and an honest map of what's next



Heart rate
working, live-demoed
~9 BPM error

10

biomarkers
rigorously tested
honestly reported

3

real bugs
found and fixed
along the way

Thank you